



INNOVATION. AUTOMATION. ANALYTICS

## PROJECT ON

Global EV Sales 2010 - 2024

Presented by:-  
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Batch:-493



# About me

- ❖ I am Sourabh Kumar Mishra ,a [B.SC](#) Computer Science Graduate .
- ❖ **Data Analytics:-**
  - ❖ Data Analysis help use to understand data to make better Decisions.
  - ❖ I want to learn data Analysis to understand data so take Smart decision instead of guessing.
  - ❖ I am interested in **data analysis and problem solving using data.**
- ❖ **LinkedIn :**
  - ❑ [www.linkedin.com/in/sourabh-kumar-mishra-930288355](http://www.linkedin.com/in/sourabh-kumar-mishra-930288355)
- ❖ **GitHub :-**
  - ❑ <https://github.com/mishragouravkumar09-boop>

# AGENDA

- ☐ Introduction
- ☐ Project Objective
- ☐ Dataset Overview
- ☐ Tools & Technologies Used
- ☐ Data Cleaning & Transformation
- ☐ Dashboard Overview
- ☐ Project explanation
- ☐ DAX Measures & Calculations
- ☐ Key Business Insights
- ☐ Challenges Faced
- ☐ Conclusion
- ☐ Q&A
- ☐ Thank You

# BUSSINESS PROBLEM.

**The global Electric Vehicle (EV) industry is growing rapidly, but EV adoption varies significantly across regions, vehicle categories, and powertrain technologies.**

**Businesses, policymakers, and analysts need a clear understanding of:**

- ☐ **Which regions are leading EV adoption,**
- ☐ **Which vehicle categories are growing fastest,**
- ☐ **How powertrain preferences are changing,**
- ☐ **How EV trends evolve over time.**

# INTRODUCTION

**Electric Vehicles (EVs) are transforming the global transportation industry by promoting cleaner and more sustainable mobility solutions. Over the last decade, EV adoption has increased rapidly due to technological advancements, government policies, and growing environmental awareness.**

**This project analyzes Global EV data from 2010 to 2024 using Microsoft Power BI to understand regional trends, powertrain growth, vehicle category performance, and overall EV market expansion through interactive dashboards and visual analytics.**

# OBJECTIVE

**The objective of this project is to analyze global Electric Vehicle (EV) adoption trends from 2010 to 2024 and derive meaningful business insights from the dataset.**

## **This project aims to:**

**study the growth of EV sales and stock across different regions,**

**compare EV adoption among vehicle categories and powertrain types,**

**identify top-performing regions and emerging markets,**

**analyze trends in BEV, PHEV, and FCEV technologies,**

**evaluate transport modes and energy consumption patterns,**

**create interactive dashboards for effective data visualization and storytelling.**

# Dataset Columns

| COLUMN 1   | COLUMN 2                                             |
|------------|------------------------------------------------------|
| Region     | Country or geographic region of EV data              |
| Category   | Vehicle type such as cars, buses, trucks, etc.       |
| Parameter  | Metrics like sales, stock, energy use, registrations |
| Mode       | Transport mode such as road or rail                  |
| Powertrain | EV technology type like BEV, PHEV, FCEV              |
| Year       | Reporting year of the data                           |
| Unit       | Measurement unit such as vehicles or TWh             |
| Value      | Numerical value of the recorded metric               |

**TOTAL RECORDS**

12,654

# Tools & Technologies Used

## Microsoft Power BI Used for:

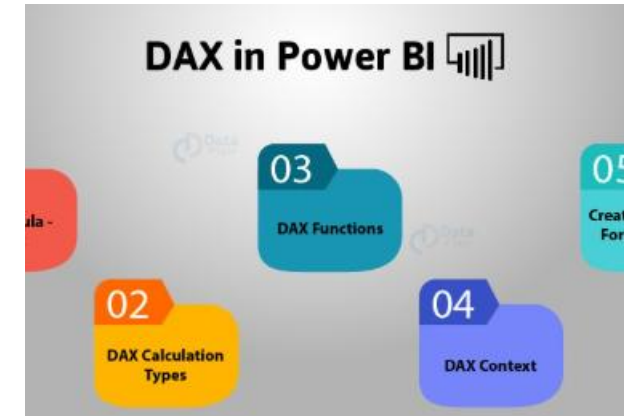
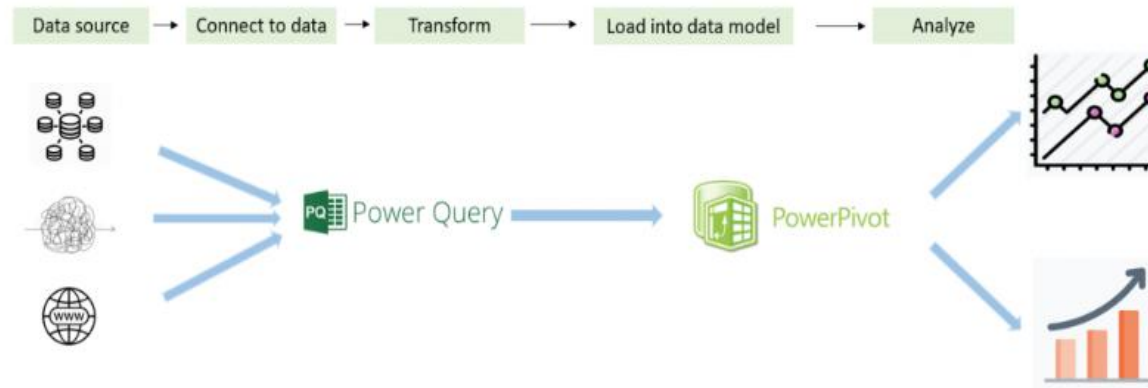
- ❑ Interactive dashboard creation
- ❑ Data visualization
- ❑ KPI analysis
- ❑ Business insights generation

## Power Query Used for:

- Data cleaning
- Data transformation
- Handling missing values
- Formatting dataset columns

## DAX (Data Analysis Expressions) Used for:

- ❖ Creating calculated measures
- ❖ KPI calculations
- ❖ Growth analysis
- ❖ Dynamic filtering and aggregations





# Data Cleaning & Transformation

## Steps Performed

Removed null and unnecessary values

Checked and corrected data types

Renamed columns for better readability

Filtered inconsistent records

Formatted year and numeric columns properly

Created calculated columns and DAX measures

Categorized regions into Developed and Developing regions

Applied aggregations and relationships for analysis

## DAX Measures Created

Total EV Sales

Total EV Stock

Year-over-Year Growth

Top N Regions

Regional Classification

Growth Percentage

# Dashboard Analysis

## Regional & Category Trends

### EV Sales Total

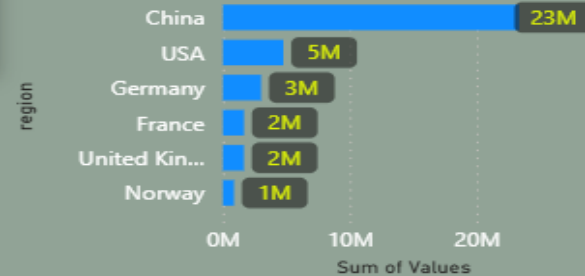
10,72,61,059.40

### EV Stock Total

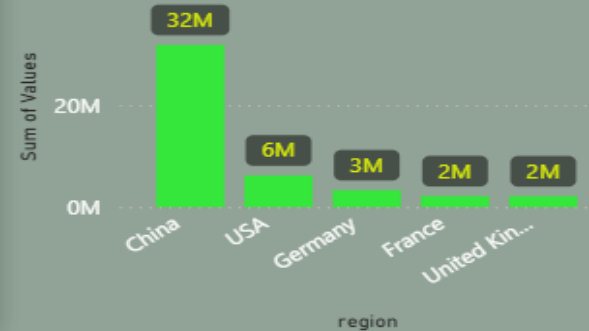
30,44,35,558.90

### Total EV Sales by region

parameter EV sales



### Top 5 regions in latest year



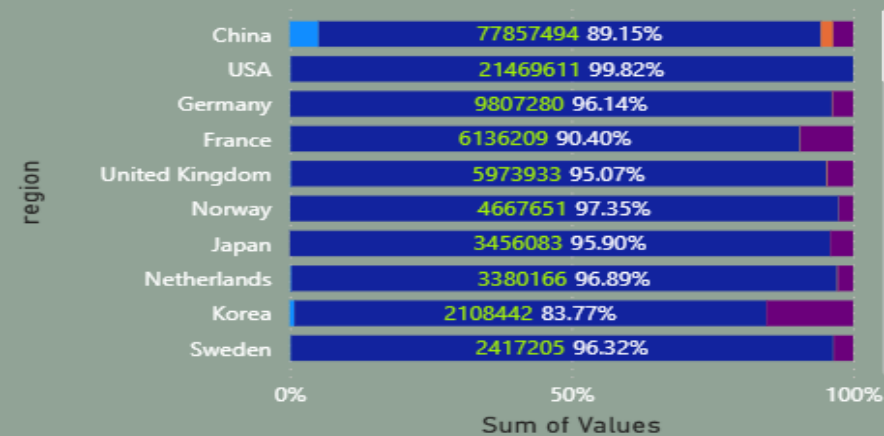
### EV Performance — Developed vs Developing Regions

Region type Developed Developing



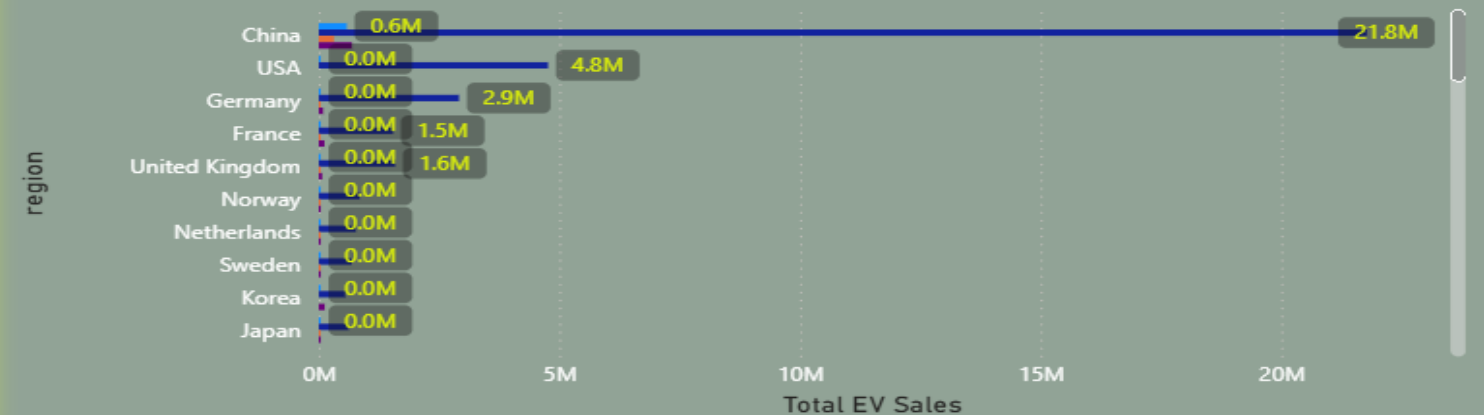
### EV Penetration by Vehicle Category and Region

vehical category Buses Cars Trucks Vans



### Region Dominance by Vehicle Category

vehical category Buses Cars Trucks Vans

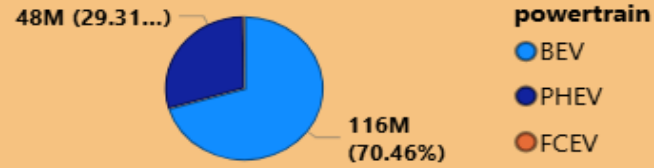


# KEY INSIGHTS

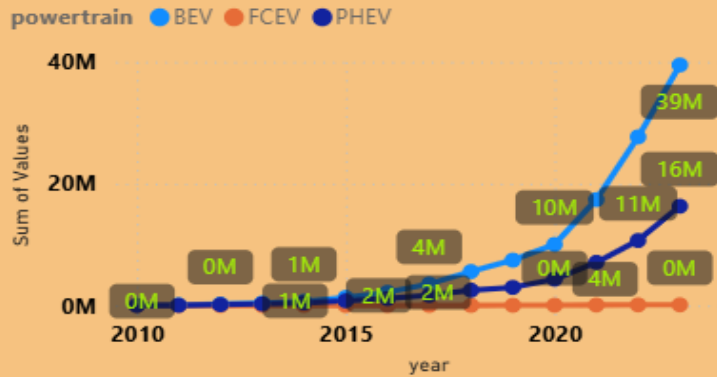
- ❑ Global EV purchases reached 418.49M, showing rapid growth.
- ❑ China leads the EV market in sales, stock, and growth.
- ❑ EV adoption increased strongly after 2020 due to demand and policies.
- ❑ 2030 and 2035 projections show major future expansion.
- ❑ Europe and China have the highest EV penetration rates.
- ❑ Passenger cars remain the dominant EV category.
- ❑ China holds the largest EV stock globally.
- ❑ Developed countries lead EV adoption, while emerging markets are growing steadily.

# Powertrain & Mode Insights

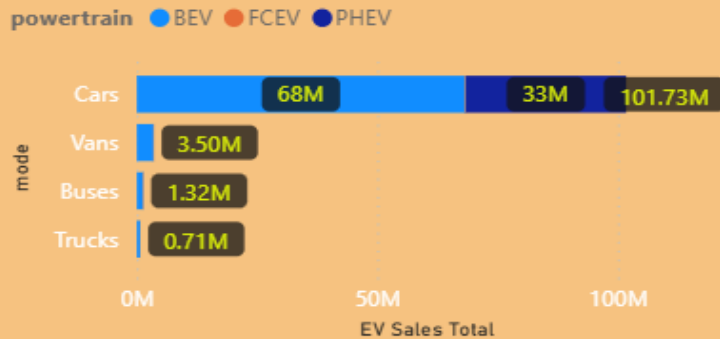
## Distribution of EVs by Powertrain



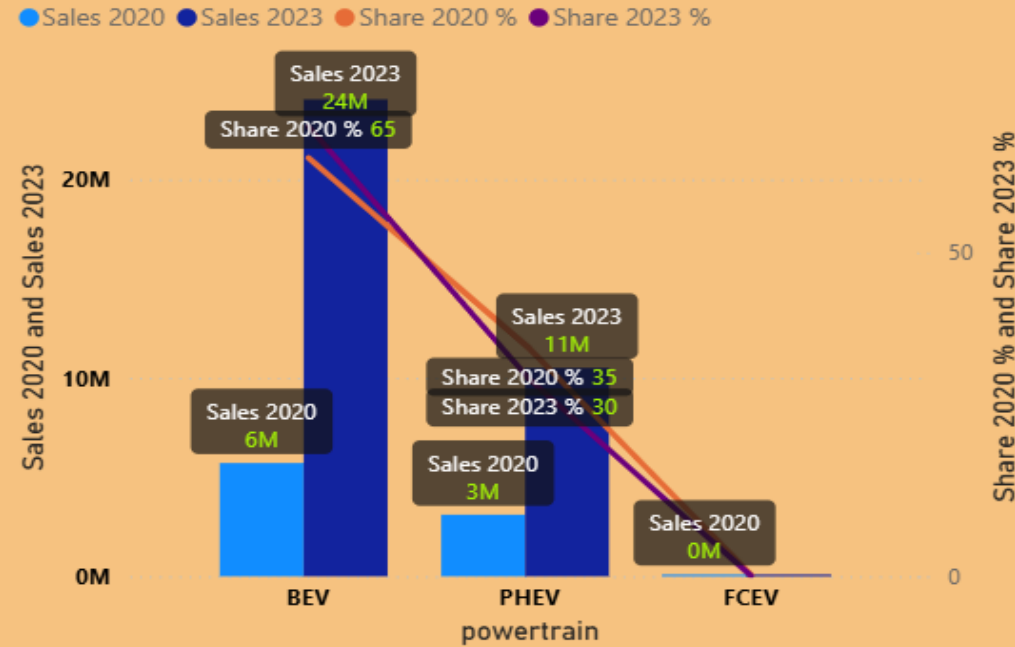
## Global Growth of Powertrain Types



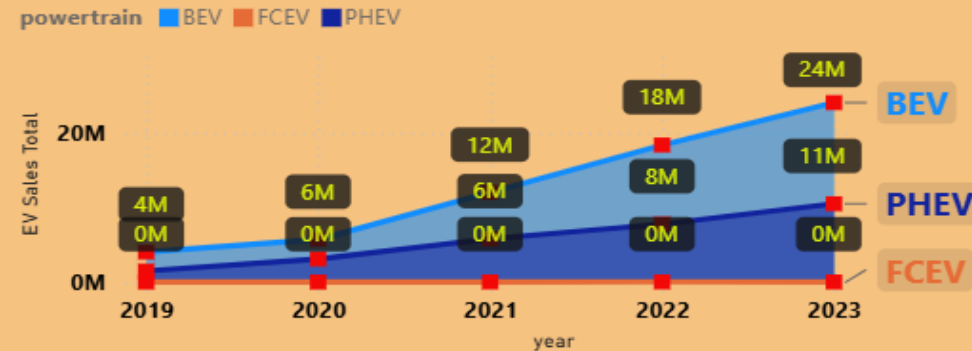
## Powertrain Adoption Across Transport Modes



## Powertrain Usage — 2020 vs 2023 Comparison



## Powertrain Trend Over Last 5 Years (2019–2023)



## Dominant Powertrain and Mode Combination by Region

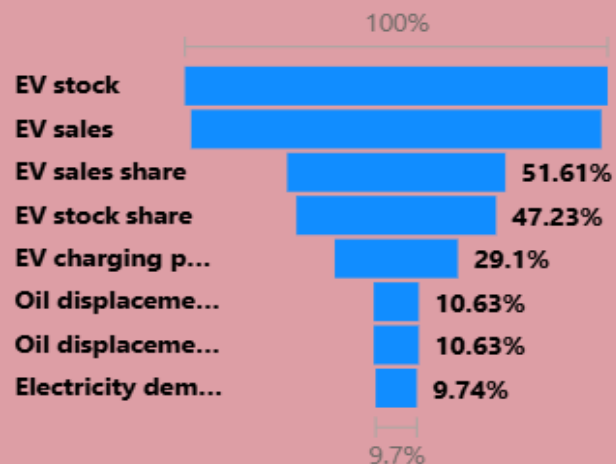
| region         | BEV       | FCEV   | PHEV |
|----------------|-----------|--------|------|
| Australia      | 447468    | 274    |      |
| Austria        | 640690    | 446    |      |
| Belgium        | 636464    | 591    | 1    |
| Brazil         | 100018    | 20     |      |
| Bulgaria       | 3834      |        |      |
| Canada         | 1586366   | 1644   |      |
| Chile          | 8295      | 7      |      |
| China          | 67804256  | 76036  | 19   |
| Colombia       | 8099      |        |      |
| Costa Rica     | 27631     |        |      |
| Croatia        | 5120      |        |      |
| Cyprus         | 1339      |        |      |
| Czech Republic | 18950     |        |      |
| Denmark        | 648091    | 1475   |      |
| Estonia        | 3210      |        |      |
| Finland        | 247844    | 8      |      |
| France         | 4472896   | 5978   | 2    |
| Germany        | 5892332   | 11424  | 4    |
| Greece         | 39148     |        |      |
| Hungary        | 21940     |        |      |
| Iceland        | 86826     | 296    |      |
| India          | 518105    | 210    |      |
| Ireland        | 57680     |        |      |
| Israel         | 258917    | 3      |      |
| Italy          | 982737    | 467    |      |
| Total          | 115965364 | 380399 | 482  |

# KEY INSIGHTS

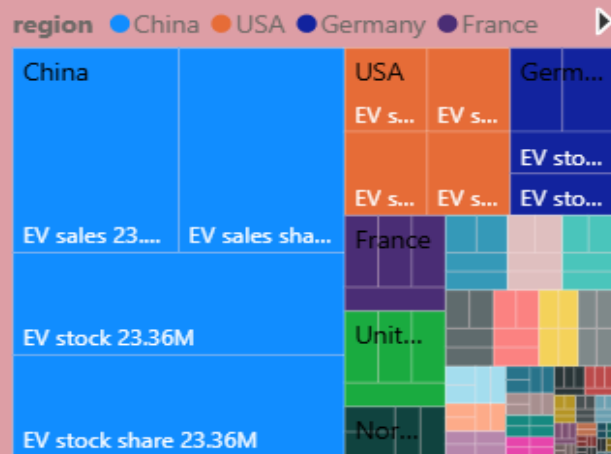
- ❑ BEVs hold the largest share in the global EV market.
- ❑ EV adoption grew rapidly after 2020.
- ❑ Passenger cars lead EV usage across transport modes.
- ❑ PHEVs are growing steadily but slower than BEVs.
- ❑ Electric buses and trucks are increasing in adoption.
- ❑ BEV passenger cars are the dominant EV combination.
- ❑ EV usage rose strongly from 2020 to 2024.
- ❑ The last 5 years show consistent EV market growth.

### Parameter & Unit Analysis

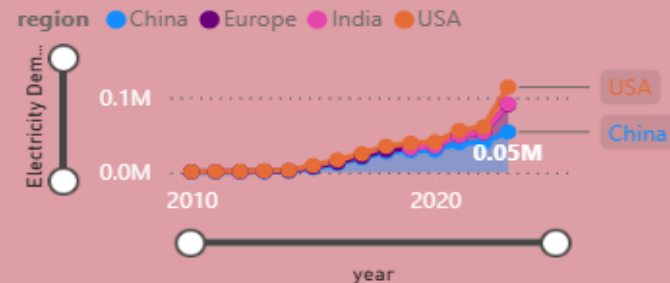
## Most Frequently Recorded EV Parameters



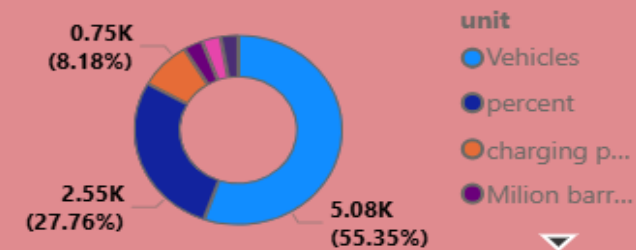
### Regional Contribution by EV Parameter



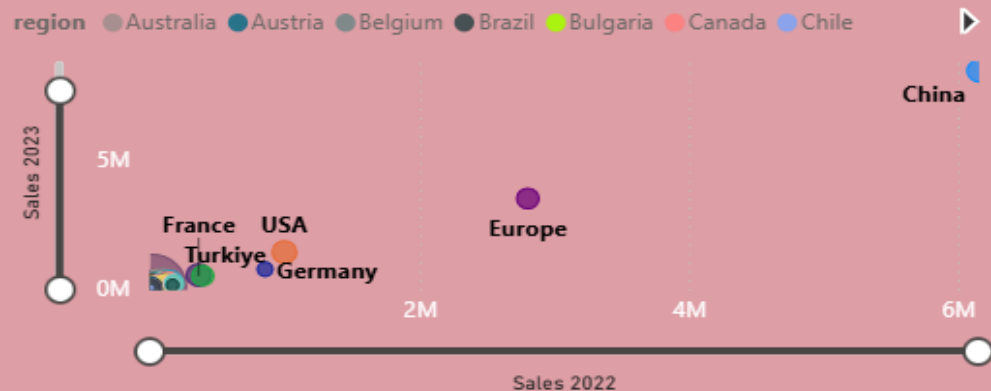
### EV Energy Consumption Trend by Region



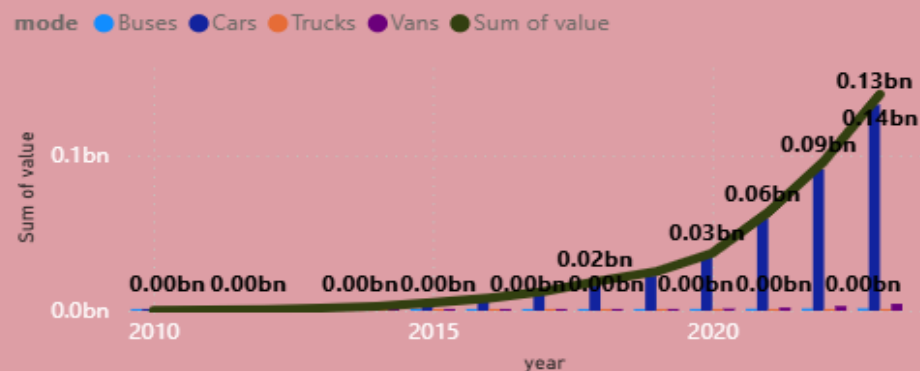
### unit Distribution Across EV Dataset



### Top Regions by EV Sales — Latest Year (2023)



## EV Stock vs New Registrations Year over Year



- region

- ☐ **Australia**
- ☐ **Austria**
- ☐ **Belgium**
- ☐ **Brazil**
- ☐ **Bulgaria**
- ☐ **Canada**
- ☐ **Chile**
- ☐ **China**
- ☐ **Colombia**
- ☐ **Costa Rica**
- ☐ **Croatia**
- ☐ **Cyprus**
- ☐ **Czech Republic**
- ☐ **Denmark**
- ☐ **Estonia**
- ☐ **Finland**
- ☐ **France**
- ☐ **Germany**
- ☐ **Greece**
- ☐ **Hungary**
- ☐ **Iceland**
- ☐ **India**
- ☐ **Indonesia**
- ☐ **Ireland**

# KEY INSIGHTS

- ❑ EV stock and sales are the most tracked EV metrics
- ❑ EV energy usage has increased steadily after 2020.
- ❑ Vehicle units dominate EV measurements globally.
- ❑ China contributes the highest EV values worldwide.
- ❑ EV sales are expected to grow rapidly by 2030–2035.
- ❑ Passenger cars remain the leading EV category.
- ❑ Charging infrastructure is expanding continuously.
- ❑ Long-term trends show strong future EV market growth.

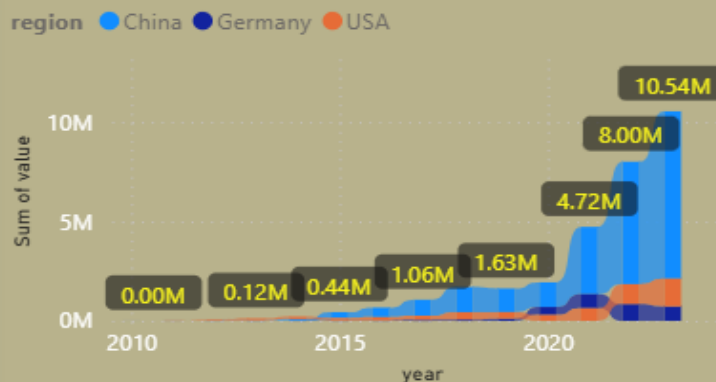


## Time Series & Comparative Analysis

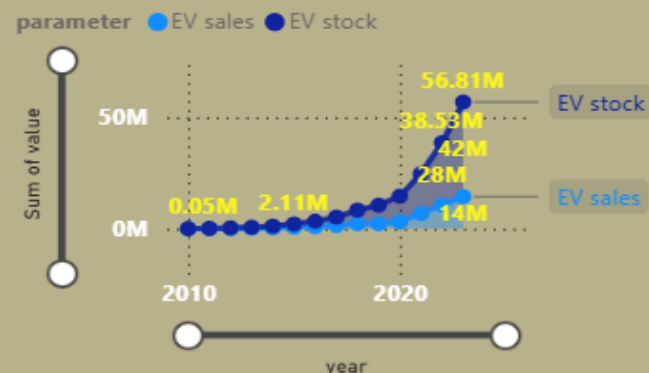
### Global EV Stock Trend 2010–2023"



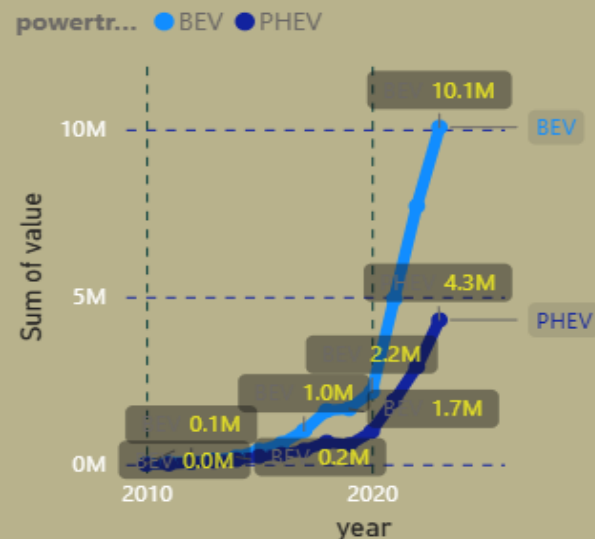
### EV Growth — Top 3 Regions 2010–2023



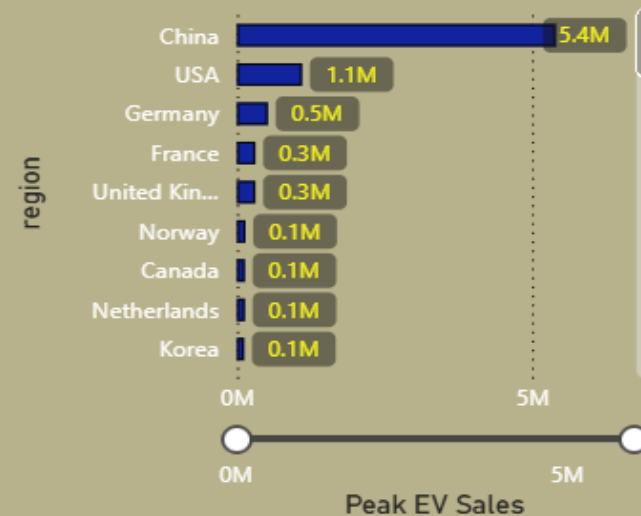
### Cumulative EV Metrics Across Years



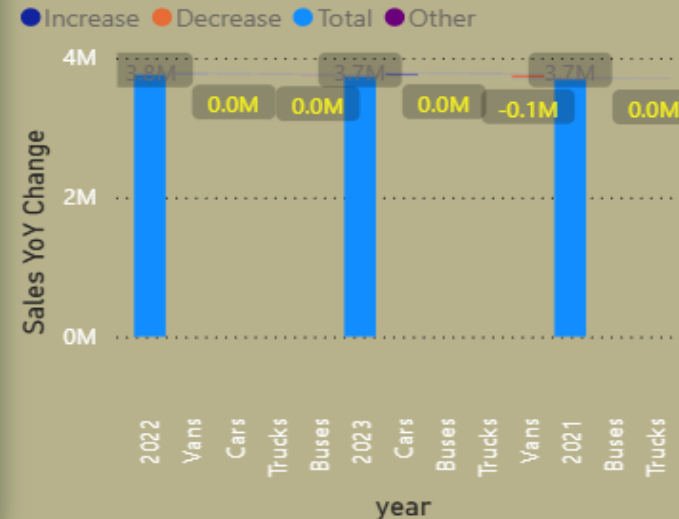
### BEV vs PHEV Sales Trend 2010–2023



### Peak EV Sales by Region



### Sales YoY Change by year and mode



region

Australia

Australia

Australia

Australia

Australia

Australia

## Australia

Australia

Australia

Australia

Australia

Australia

Australia

Australia

Austria

Austria

Austria

Austria

Austria

Austria

Austria

Austria

Austria

Austria

Total



# KEY INSIGHTS

- ❑ Global EV stock grew rapidly after 2020.
- ❑ China leads global EV stock growth.
- ❑ EV sales, stock, and infrastructure increased yearly.
- ❑ Passenger cars and buses showed the highest growth.
- ❑ BEVs are growing faster than PHEVs.
- ❑ Peak EV sales occurred between 2021–2023.
- ❑ Government support accelerated EV adoption after 2020.
- ❑ Future trends show strong long-term EV market growth.

# Business Insights

- ❑ Global EV adoption increased rapidly from 2010–2024
- ❑ China is the global leader in EV sales and stock
- ❑ BEVs dominate the EV market compared to PHEVs and FCEVs
- ❑ Developed regions show higher EV adoption rates Developing countries like India show strong future growth potential
- ❑ Passenger cars contribute the highest EV sales globally
- ❑ Road transport is the dominant EV transport mode
- ❑ EV growth accelerated significantly after 2020
- ❑ Energy consumption related to EVs is increasing steadily
- ❑ Microsoft Power BI dashboards improved data analysis and business decision-making

# Recommendations

- **Governments should expand EV charging infrastructure**
- **Increase investment in Battery Electric Vehicle (BEV) technology**
- **Encourage EV adoption through subsidies and incentives**
- **Focus on developing EV infrastructure in emerging markets**
- **Promote sustainable transportation policies globally**
- **Support public transport electrification such as EV buses**
- **Improve renewable energy integration for EV charging**
- **EV manufacturers should target developing countries for future growth**
- **Increase awareness about environmental benefits of EVs**
- **Use data analytics tools like Microsoft Power BI for better market analysis and decision-making**

# Challenges Faced

**Handling large and complex EV datasets**

**Managing aggregation and filtering issues**

**missing and inconsistent data**

**Creating accurate DAX measures and calculations**

**Designing interactive and visually appealing dashboards**

**Handling null values replacing , Duplicates**

# CONCLUSION

- ❑ Global EV adoption has increased significantly from 2010 to 2024
- ❑ China emerged as the leading region in EV sales and stock
- ❑ BEVs became the most dominant powertrain technology globally
- ❑ Passenger cars contributed the highest share of EV adoption
- ❑ EV growth accelerated rapidly after 2020 due to policy support and technological advancements
- ❑ Developed regions currently lead the EV market, while developing countries show strong future potential
- ❑ Microsoft Power BI helped transform raw EV data into interactive dashboards and meaningful business insights
- ❑ The project improved understanding of EV trends, regional performance, and data-driven decision-making

Q&A

THANK YOU

